

No. of Printed Pages : 4
Roll No.

221063A

**6th Sem/
ECE, ECE (For Speech & Hearing Impaired)
Subject : Microwave and Radar Engineering**

Time : 3 Hrs.

M.M. : 60

SECTION-A

Note: Multiple choice questions. All questions are compulsory (6x1=6)

- Q.1 The dominant mode in a rectangular waveguide is
a) TE₁₀ b) TE₁₁
c) TM₀₁ d) TM₁₁
- Q.2 Which one of the following can be used for amplification of microwave energy
a) TWT b) Magnetron
c) Reflex Klystron d) Gunn diode
- Q.3 Magic tee is nothing but.
a) A modification of E-Plane tee
b) A modification of H-Plane tee
c) Two E-plane tee connected in parallel
d) A combination of E plane and H plane tee
- Q.4 Microwaves are used for
a) Remote sensing b) Medical applications
c) Communication d) All above

Q.5 Klystron operates on the principle of

- a) Velocity modulation
- b) Amplitude modulation
- c) Pulse Modulation
- d) Frequency modulation

Q.6 Radar is used for :

- a) Ground mapping only
- b) Airport surveillance only
- c) Weather forecasting only
- d) All of above

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (6x1=6)

Q.7 The full form of TEM is _____.

Q.8 What is frequency range of microwave _____.

Q.9 Gunn diode is a Transferred electron Device. (T/F)

Q.10 Full form of PPI is _____.

Q.11 Frequency range of S band is _____.

Q.12 Echo is a reflected signal. (T/F)

SECTION-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Draw the block diagram of a basic radar system.

- Q.14 List any four advantages of a waveguide.
- Q.15 Explain the advantages of microwave over low frequency signals.
- Q.16 List the various modes of propagation in a waveguide.
- Q.17 Write the radar range equation.
- Q.18 Write the basic principle of operation of TWT
- Q.19 Write any four applications of Microwaves.
- Q.20 Draw and explain the working of a variable attenuator.
- Q.21 List any four applications of IMPATT.
- Q.22 Explain any four applications of magic tee.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

- Q.23 Explain the basic principle of MTI radar system with the help of block diagram.
- Q.24 Write short notes on :
a) Circular waveguide b) Slotted Section
- Q.25 What is a reflex klystron? Explain its working with the help of a neat sketch.

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